

IP Address Management (IPAM) Server Configuration

After completing the core infrastructure setup, an IP Address Management (IPAM) server was deployed on Hyper-V to centrally monitor and manage IP address allocation across the enterprise network environment.

1. Initial Server Configuration

The IPAM server was prepared with basic system settings before enabling IPAM features.

Configuration Procedure

1. Open **Server Manager**.
2. Configure the following from **Local Server**:
 - Computer Name
 - Time Zone
 - Static IP Address

Server Configuration Details

Parameter	Value
Hostname	IPAM
Time Zone	Dhaka
IP Address	192.168.1.15
Subnet Mask	255.255.255.0
Default Gateway	pfSense LAN Gateway
DNS Server	192.168.1.10

Result

The IPAM server was successfully configured with hostname, timezone, and static IP address.

2. IPAM Feature Installation

IPAM feature was installed and enabled on Windows Server 2022.

Installation Procedure

1. Open **Server Manager**.

2. Click **Add Roles and Features**.
3. Select **Role-based or feature-based installation**.
4. Select server (**IPAM**).
5. Navigate to **Features** section.
6. Select **IP Address Management (IPAM) Server** feature.
7. Click **Install**.
8. Complete installation.

Result

The IPAM feature was successfully installed on the server.

3. IPAM Provisioning Configuration

After installation, IPAM provisioning was initialized to enable discovery and management of network infrastructure.

Configuration Procedure

1. Open **PowerShell as Administrator**.
 2. Run IPAM provisioning command:
- ```
Invoke-IPAMGPOProvisioning
```
3. Create and configure IPAM GPO settings.
  4. Open **Group Policy Management Console (GPMC)**.
  5. Create a new GPO named:

IPAM

6. Run Group Policy update:

```
gpupdate /force
```

## Result

IPAM provisioning was successfully completed and Group Policy Objects were applied across the domain.

# 4. IPAM Server Discovery and Configuration

After provisioning, IPAM began discovering network infrastructure components including DNS, DHCP, and domain controllers.

## Configuration Procedure

1. Open **Server Manager**.
2. Navigate to **IPAM** console.
3. Configure server discovery scope.
4. Add managed servers (DC01 and network services).
5. Complete IPAM access configuration.
6. Wait for synchronization and data collection.

## Result

IPAM successfully discovered and began managing network infrastructure services within the domain.

## Final Result

The IP Address Management (IPAM) server was successfully deployed and configured, providing centralized IP tracking and management for the enterprise infrastructure environment.

The system achieved:

- Centralized IP address monitoring
- DHCP scope visibility
- DNS integration monitoring
- Active Directory integration
- Group Policy-based IPAM provisioning
- Automated infrastructure discovery